

Enterprise Infrastructure Plan

TechLine Solutions

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Company Profile

Company Name: TechLine Solutions

Nature of Business: A software development startup specializing in custom web and mobile applications for small to medium-sized businesses.

Company Vision: To become a leading software development company known for innovative, reliable, and affordable technology solutions across the Philippines and Southeast Asia.

Company Mission: To empower businesses through accessible software solutions, while fostering a culture of continuous learning, innovation, and technical excellence among our team.

Office Location: 4th Floor, TechHub Business Center, Calamba City, Laguna, Philippines (fictional)

Organizational Structure:

- CEO / Managing Director
 - IT Department – Systems, infrastructure, and technical support
 - Human Resources Department – Recruitment, employee relations, and administration
 - Finance Department – Accounting, payroll, and budgeting
 - Sales Department – Client acquisition and account management

Employee Distribution:

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	14	HR, Finance, Sales	Standard workstations for employees performing office-based, non-mobile tasks
HW-002	Laptop	6	IT	Portable workstations for IT staff who need to move between departments for support and maintenance

HW-003	Server (Tower)	1	IT	Hosts internal applications, file storage, and acts as the domain/network server
HW-004	Router	1	IT (Network Closet)	Manages internet traffic and connects the office network to the ISP
HW-005	Switch	2	IT (Network Closet)	Distributes wired network connections to all desktops and devices across departments
HW-006	Printer (Network)	2	Shared (Finance/Sales, HR/IT)	Shared printing for documents, contracts, and reports across departments
HW-007	UPS (Uninterruptible Power Supply)	3	Server Room, Network Closet	Protects the server and networking equipment from power interruptions and outages
HW-008	Wireless Access Point	2	IT (Network Closet)	Provides Wi-Fi coverage for laptops and mobile devices throughout the office
HW-009	NAS Storage	1	IT	Centralized file storage and backup for shared company documents
HW-010	External Backup Drive	2	IT	Offline backup copies of critical company data, separate from the NAS
HW-011	Monitor	20	All Departments	Display units for all desktop and laptop workstations

Justification Summary:

Desktop computers are assigned to HR, Finance, and Sales since their work is primarily office-based. IT staff are given laptops instead, since their role often requires mobility for troubleshooting across departments. One server is sufficient at this company size to handle file sharing and internal hosting needs, supported by a NAS and external drive for redundancy in backups. Two switches and two access points ensure reliable wired and wireless coverage for a single-floor office of 20 employees, while a shared printer setup avoids unnecessary equipment costs. UPS units are placed specifically at the server and network closet since those systems are most critical to keep running during a power interruption.

Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume License (per device)	Primary operating system for all employee desktops and laptops, chosen for business features like BitLocker and domain join support
Ubuntu Server	22.04 LTS	Free / Open Source	Operating system for the company server, used for file sharing, hosting internal tools, and backups
Microsoft Office	365 Business	Subscription (per user)	Word processing, spreadsheets, and presentations for daily administrative and business tasks
Visual Studio Code	Latest	Free / Open Source	Code editor for the IT and development team to write and maintain software

Git	Latest	Free / Open Source	Version control system used by the IT team to track changes in code and configuration files
GitHub Desktop	Latest	Free	Simplified visual interface for managing Git repositories, used by IT staff
VirtualBox	Latest	Free / Open Source	Virtualization software used by IT for testing environments and running virtual machines
Google Chrome	Latest	Free	Primary web browser for all employees across departments
Microsoft Defender	Built-in (Windows 11)	Included with Windows	Baseline antivirus and endpoint protection for all company workstations
AnyDesk	Latest	Free (Business tier optional)	Remote desktop tool used by IT for troubleshooting employee workstations remotely
7-Zip	Latest	Free / Open Source	File compression and archiving tool for handling large files and backups

Justification Summary:

Windows 11 Pro is standard across employee workstations due to its business-grade security features, while Ubuntu Server runs the company server since it is stable, free, and well-suited for hosting and file services. Microsoft Office covers everyday administrative needs across HR, Finance, and Sales. Git, GitHub Desktop, VS Code, and VirtualBox are IT/development-specific tools needed for building, testing, and maintaining internal systems. Chrome is the standard browser for general use, while Microsoft Defender provides baseline protection without added licensing cost. AnyDesk allows the IT department to support employees remotely without needing to physically visit every workstation, and 7-Zip supports basic file management and compression needs across the company.

Enterprise Network Inventory

Asset ID	Networking Equipment	Quantity	Purpose
NW-001	ISP Modem	1	Provides the internet connection from the Internet Service Provider into the office network
NW-002	Router	1	Directs traffic between the internal network and the internet, and manages IP addressing
NW-003	Firewall	1	Filters incoming and outgoing traffic to protect the internal network from external threats
NW-004	Switch	2	Connects wired devices (desktops, printers, server) within each department to the network
NW-005	Wireless Access Point	2	Extends Wi-Fi coverage for laptops and mobile devices across the office floor
NW-006	Patch Panel	1	Organizes and terminates all CAT6 cable runs from workstations into the network closet
NW-007	CAT6 Cables	30	Wired connections between workstations, switches, and the patch panel
NW-008	RJ45 Connectors	60	Used to terminate CAT6 cable ends for network connections

Justification Summary:

The ISP modem and router form the entry point of internet connectivity into the office, while the firewall adds a dedicated layer of protection against external threats before traffic reaches internal devices. Two switches are used to comfortably support wired connections across all four departments without overloading a single switch. Two wireless access points ensure reliable Wi-Fi coverage across the single-floor office, especially for IT staff using laptops. The patch panel keeps cable management organized in the network closet, and CAT6 cabling with RJ45 connectors is used since it reliably supports the gigabit speeds needed for daily business operations.

System Administration Roles

Helpdesk Technician

Responsibilities: Provides first-line technical support to end-users, troubleshooting hardware, software, and connectivity issues. Manages support tickets, resets passwords, installs and configures software, and escalates complex problems to higher-level IT staff. A helpdesk technician also supports operating systems, productivity software, and common business applications, and helps manage devices through tools like Microsoft Intune.

Skills: Strong troubleshooting ability, clear communication, patience with frustrated users, and basic knowledge of Windows, Active Directory, and Microsoft 365. Basic scripting in PowerShell or Bash is a valuable plus for automating routine tasks like password resets or software installations.

Common Tools: Ticketing systems such as ServiceNow, Zendesk, or Freshdesk, along with Remote Desktop, Active Directory, and Microsoft 365 admin tools.

Certifications: CompTIA A+ is a widely recognized entry-level certification covering hardware, software troubleshooting, networking, and security fundamentals. The Cisco Certified Support Technician (CCST) is another option for entry-level IT roles.

Network Administrator

Responsibilities: Monitors network performance, troubleshoots and analyzes issues, and monitors the state of routers, firewalls, and network hardware. Enforces security by managing firewalls and access permissions, and handles hardware and software management including installing routers, setting up operating systems, and performing upgrades.

Skills: Proficiency in networking protocols like TCP/IP, DNS, DHCP, and VPN, strong organizational skills, and security awareness to protect organizational data. Increasingly, network administrators also need cloud networking experience with platforms like AWS, Azure, or Google Cloud, since they now maintain both on-premise and cloud networks.

Common Tools: Network monitoring software such as SolarWinds, Nagios, and PRTG Network Monitor to track network performance and detect issues.

Certifications: Cisco Certified Network Associate (CCNA), CompTIA Network+, and Certified Information Systems Security Professional (CISSP).

Linux System Administrator

Responsibilities: Basic system management, user administration, storage management, configuration, troubleshooting, and ensuring security on a Linux server.

Skills: Ability to use essential tools for handling files, directories, and command-line environments, and operate running systems including booting into different run levels, identifying processes, and controlling services. Configuring local storage using partitions and logical volumes, and creating file systems with attributes like permissions, encryption, and access control lists.

Common Tools: Shell/command-line tools, package managers, and container management tools like Podman and Skopeo for handling container images and services.

Certifications: Red Hat Certified System Administrator (RHCSA), earned by passing the performance-based EX200 exam, validates core Linux system administration skills.

Cloud Administrator

Responsibilities: Keeping cloud infrastructure usable, secure, and cost-effective, which includes provisioning virtual machines, managing identities, adjusting storage, configuring networks, monitoring health, and responding to alerts.

Skills: A substantial understanding of Linux and Windows, since most cloud environments operate on one or both, along with comfort using the command line, scripting, and system administration. Scripting knowledge in languages like Bash, Python, and PowerShell to automate cloud tasks.

Common Tools: AWS Management Console, Microsoft Azure Portal, and Google Cloud Console, along with tools like Terraform and Ansible for infrastructure as code, and monitoring tools like Amazon CloudWatch and Azure Monitor.

Certifications: AWS Certified Solutions Architect and Microsoft Certified: Azure Administrator Associate.

How These Roles Work Together

In a company like TechLine Solutions, these four roles form a layered support structure that keeps IT operations running smoothly. The Helpdesk Technician acts as the first point of contact, resolving everyday issues and escalating anything beyond basic troubleshooting. When a problem involves the network itself, such as connectivity issues or hardware failures, it gets passed to the Network Administrator, who manages the routers, switches, and firewall. If the company server needs configuration, patching, or troubleshooting at the operating system level, the Linux System Administrator steps in to maintain that Ubuntu Server environment. As the company grows and potentially adopts cloud-based tools for backups, remote access, or scalable storage, the Cloud Administrator manages those resources, working closely with the Network Administrator on connectivity and the Linux Administrator on server configuration. Together, these roles form a chain of responsibility, from the individual employee's desk to the company's core infrastructure, ensuring that both daily user needs and long-term system stability are covered.

Infrastructure Recommendations

- **Internet Provider**

For a 20-employee company handling daily cloud-based tools, email, and file sharing, a fiber internet connection with at least 500 Mbps is recommended over standard DSL or cable, since fiber offers more reliable upload speeds, which matters for uploading files and hosting internal services. A business-grade plan with a Service Level Agreement (SLA) is preferred over a residential plan, since SLAs guarantee faster support response times if the connection goes down.

- **Server Specifications**

The company server should run Ubuntu Server LTS for stability and long-term support, with enough RAM (at least 16GB) and storage to comfortably handle file sharing, internal hosting, and backups for 20 users without slowing down. A RAID configuration (such as RAID 1 or RAID 5) is recommended for the server's storage drives, so that data is mirrored or distributed across multiple disks, reducing the risk of total data loss if a single drive fails.

- **Backup Strategy**

A 3-2-1 backup approach is recommended: 3 copies of important data, stored on 2 different types of media (such as the NAS and an external backup drive), with 1 copy kept offsite or in the cloud. Backups should run automatically on a daily schedule for critical files, with weekly full backups of the entire server.

- **Security Recommendations**

The network should be segmented so that guest or non-essential devices are not on the same network as the core business systems. The firewall should be configured to block unnecessary inbound traffic by default, and only allow the specific ports and services the company actually needs. Regular software and firmware updates should be applied to routers, switches, and servers to patch known vulnerabilities.

- **Antivirus**

Since the company uses Windows 11 Pro on employee workstations, Microsoft Defender provides a solid baseline of protection without added licensing cost, especially when paired with good user habits and regular updates. For a company handling sensitive Finance and HR data, upgrading to a managed endpoint protection solution as the company grows would add centralized monitoring and faster threat response.

- **Password Policy**

Employees should be required to use passwords that are at least 12 characters long, combining uppercase, lowercase, numbers, and symbols. Passwords should be changed periodically (e.g. every 90 days) for accounts with access to sensitive systems, and multi-factor authentication (MFA) should be enabled wherever possible, especially for email and remote access accounts.

- **Expansion Plan**

As TechLine Solutions grows beyond 20 employees, the network should be able to scale by adding additional switches and access points without redesigning the entire topology. The server setup should also be reviewed periodically, since a growing company may eventually need a dedicated backup server separate from the primary file server, or a move toward hybrid cloud storage for better scalability and remote access.

Personal Reflection

Working on this Enterprise Infrastructure Plan for TechLine Solutions gave me a much clearer picture of what actually goes into setting up a company's IT environment before a single computer is even purchased. Coming into this project, I assumed that being a System Administrator was mostly about fixing things when they broke or managing a network that already exists. This project showed me that a large part of the role happens before deployment, translating a business's size and departmental needs into a concrete list of hardware, software, and network requirements that someone could actually act on.

The most challenging part of this project was the Enterprise Network Diagram. Even after understanding conceptually how a network flows from the internet down to individual departments, actually representing that visually in **Draw.io** took several attempts to get right. I had to learn how to use proper networking symbols, correctly label the connections between devices, and make sure the layout was clean enough to read at a glance. Comparing my early drafts to more detailed examples from classmates also helped me realize there's a wide range of how much detail a network diagram can include, from a simple functional layout to one with specific device models and VLAN segmentation, and I had to make a judgment call on what level of detail made sense given my time and the actual project requirements.

Planning before deployment is important because mistakes made on paper are far cheaper to fix than mistakes made after equipment is already purchased and installed. If a company buys the wrong number of switches, or fails to account for a department's specific needs, that becomes an expensive problem to correct later. Proper planning also forces you to think about scalability and security from the very beginning, rather than trying to bolt them on after the fact. Writing out a hardware and software inventory, in particular, made me realize how many small decisions (like how many switches are actually needed for a specific number of devices) require actual reasoning rather than guesswork.

This project will help me become a better System Administrator because it pushed me to think at a system level rather than just focusing on individual devices or issues. Researching the different IT roles, like Network Administrators and Cloud Administrators, also gave me a clearer sense of how a real IT department is structured and how these responsibilities are divided in practice. Going forward, I feel more confident approaching infrastructure problems methodically, starting with requirements and constraints before jumping into solutions, which feels like a foundational habit for anyone working in system administration.